

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			$\frac{0.001}{0.500} \times 100 = 0.20 \%$ award (1) if one error made e.g. failure to convert mg to g		2		2	1	2
	(b)	(i)		it is the number of moles of ascorbic acid used that is important / concentration of acid is not important			1	1		1
		(ii)		heated to ensure that all the acid / the tablet dissolved		1		1		1
	(c)			total volume of titres = $26.73 \times 3 = 80.20 \text{ cm}^3$ / third titre is 26.80 cm^3 (1) so final reading must be 27.00 cm^3 (1) do not accept 26.99 cm^3		1				
							1	2	2	2
	(d)			volume of sodium hydroxide added would be less (1) since fewer moles of acid present / calculated mass of acid in tablet would be less (1) award (2) for volume NaOH added more since OH groups in ascorbic acid oxidised to acid		1	1	2		2
	(e)			moles NaOH = $2.67(3) \times 10^{-3}$ which is equal to moles acid (1) mass acid = 0.4704 g (1) percentage acid = 94.1 (1)						
						3		3	2	

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	(f)	(i)		[H ⁺] = 0.02 (1) pH = 1.7 (1) award (1) for pH = 2		2		2	2	
		(ii)		oxidation state Mn changes from IV to II therefore it is reduced (1) oxidation state Cl changes from –1 to 0 therefore it is oxidised (1)		2		2		
				Question 9 total	0	12	3	15	7	8